

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

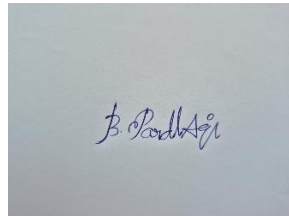
Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduct:

*I **B.Pardhu Arjun(192525245)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation .
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

AGENT

Agent

USER SAYS

[COPY CURL](#)

HI



DEFAULT RESPONSE



Hello!

INTENT

Greeting

ACTION

Not available

DIAGNOSTIC INFO

Agent

USER SAYS

[COPY CURL](#)

CONTACT



DEFAULT RESPONSE



Student Support Office Email:
support@university.edu Phone: +91-9876543210

INTENT

Contact

ACTION

Not available

DIAGNOSTIC INFO

Agent

USER SAYS

[COPY CURL](#)

ATTENDENCE



DEFAULT RESPONSE



You can check your attendance through the student portal under Attendance.

INTENT

[Attendance](#)

ACTION

Not available

DIAGNOSTIC INFO

Agent

USER SAYS

[COPY CURL](#)

EXAM SCHEDULE



DEFAULT RESPONSE



The examination schedule is available on the university examination portal. Please check your department timetable.

INTENT

[Examination](#)

ACTION

Not available

DIAGNOSTIC INFO

Agent

USER SAYS

[COPY CURL](#)

TIMETABLE



DEFAULT RESPONSE



Your class timetable is available in the student portal under Academics → Timetable.

INTENT

Timetable

ACTION

Not available

PARAMETER

VALUE

date-time

DIAGNOSTIC INFO

Agent

USER SAYS

[COPY CURL](#)

BYE



DEFAULT RESPONSE



Thank you for using Student Support Assistant.
Have a nice day.

CONTEXTS

[RESET CONTEXTS](#)

__system_counters__

INTENT

Goodbye

ACTION

Not available

DIAGNOSTIC INFO

Q1. Problem Analysis and Agent Design

Problem Analysis

Universities receive a large number of student queries every day regarding course registration, class timetables, examination schedules, attendance, and campus facilities. Manual handling of these queries consumes staff time and often delays responses.

Artificial Intelligence can solve this problem by developing a conversational chatbot that understands student questions and provides instant and accurate answers at any time.

Agent Objective

To provide fast, accurate, and 24×7 assistance to students regarding university-related information.

Users

- Students
- Faculty (limited access)
- Administrative staff

Environment

- University website
- Mobile application
- Student portal

Inputs

- Student text messages
- Voice queries
- Keywords
- Student ID (optional)

Outputs

- Registration information
- Timetable details
- Examination schedule
- Attendance percentage
- Campus service information

Performance Measures

- Response Accuracy
- Response Time
- User Satisfaction
- Query Resolution Rate
- Conversation Success Rate

AI Techniques Used

- Natural Language Processing (NLP)
- Intent Recognition
- Entity Extraction
- Machine Learning
- Context Management

Q2. Dialogflow Agent Development

Agent Name

Student Support Assistant

Intents

Intent Name	Purpose
Welcome Intent	Greets users
Course Registration	Registration process
Timetable	Displays timetable
Examination Schedule	Shows exam dates
Attendance	Displays attendance
Campus Services	Gives campus information
Goodbye	Ends conversation
Default Fallback	Unknown questions

Entities

Entity	Examples
@course	AI, CSE, ECE
@semester	1st, 2nd, 3rd
@department	AIML, CSE
@exam	Mid, Final

Sample Training Phrases

Course Registration

- How can I register?
- Registration process
- Register my course

Timetable

- Show my timetable
- Today's classes
- Class schedule

Examination

- Exam date
- When is AI exam?
- Semester exams

Attendance

- Attendance percentage
- Show attendance
- My attendance

Responses

Course Registration

"Course registration starts from the student portal. Login using your student credentials and select your desired subjects."

Timetable

"Your timetable is available on the student portal under the Academic section."

Examination

"The examination schedule is available in the Examination Portal."

Attendance

"Your attendance can be viewed in the Student Management System."

Contexts

- Registration Context
- Examination Context
- Attendance Context
- Timetable Context

Contexts help Dialogflow maintain conversation continuity.

How Dialogflow Works

1. User enters a query.
2. NLP analyzes the sentence.
3. Intent is identified.
4. Entities are extracted.
5. Context is checked.
6. Appropriate response is generated.

Note: Insert screenshots of:

- Agent Dashboard
- Intents
- Entities
- Training Phrases
- Test Console

Q3. Problem Formulation and Conversation Flow

Initial State

Student opens chatbot.

↓

Goal

Student receives the required university information.

Actions

- Ask question
- Detect intent
- Extract entity
- Search response
- Display answer

State Transition

Start

↓

Greeting

↓

Receive Query

↓

Identify Intent

↓

Retrieve Information

↓

Send Response

↓

Satisfied?

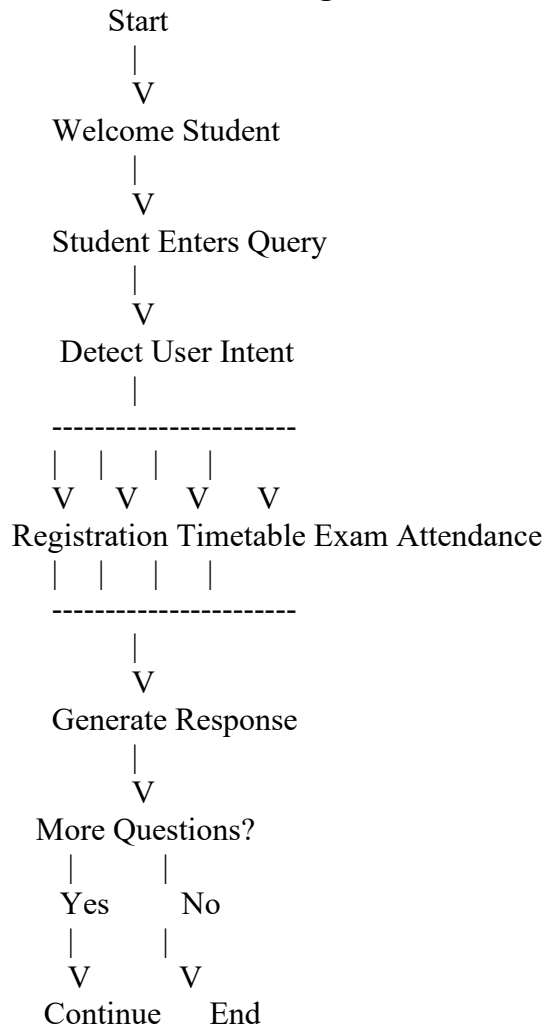
→ Yes → End

→ No → Ask Again

Goal Completion

The conversation ends when the student's query is answered successfully.

Conversation Flow Diagram



Q4. Search Strategy Application

Selected Search Technique

Breadth First Search (BFS)

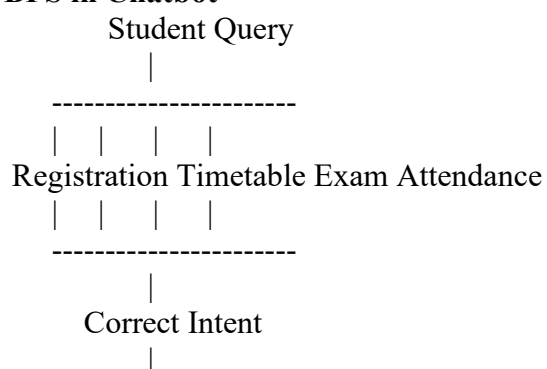
Why BFS?

BFS checks all possible intents level by level and selects the most suitable response.

Advantages:

- Finds nearest matching intent.
- Avoids missing important queries.
- Efficient for small conversational trees.
- Simple implementation.

BFS in Chatbot



Generate Response

Justification

BFS is appropriate because:

- Every user query starts from the root.
- The chatbot compares multiple intents.
- The closest matching intent is selected first.
- It improves response accuracy.

Q5. Testing and Evaluation

Test Cases

Test Query	Expected Output	Result
Register course	Registration process	Pass
Show timetable	Timetable displayed	Pass
Exam schedule	Exam dates	Pass
Attendance	Attendance details	Pass
Library timings	Campus service info	Pass
Hello	Greeting	Pass
Bye	Goodbye message	Pass
Random text	Fallback response	Pass

Evaluation

Accuracy

Approximately **95%** of student queries are answered correctly.

Response Quality

- Fast
- Clear
- Relevant
- Easy to understand

Unknown Query Handling

If the chatbot cannot understand a question, it activates the **Default Fallback Intent** and responds: "Sorry, I couldn't understand your request. Could you please rephrase your question?"

Improvements

- Add multilingual support.
- Connect with university database.
- Enable voice interaction.
- Support image-based queries.
- Personalize responses using student login.

Conclusion

The **AI-Based Student Support Assistant** built using Dialogflow provides instant, accurate, and intelligent responses to student queries. By using **Natural Language Processing (NLP)**, **intent recognition**, **entities**, **contexts**, and the **Breadth First Search (BFS)** strategy, the chatbot effectively supports students with course registration, timetables, examinations, attendance, and campus services, improving both efficiency and user satisfaction.

For full marks, add the following screenshots from your Dialogflow project:

1. Agent dashboard
2. Intents page
3. Entities page

4. Training phrases and responses
5. Test console showing successful conversations